Exp3: For A SET of Training DATA Examples store in a CSV file Demonstrate Data Preprocessing

1. Getting the data set.
2. Importing libraries.
3. Importing Datasets.
4. Finding Missing Data.
5. Encoding Categorial.
6. Splitting dataset into training and test rest.
7. Feature scaling.

Aim :- To demonstrate data preprocessing for a given Dataset is stored in a csv file and the preprocessing steps include loading the data, handling, missing values & applying feature scaling.

1. Getting the Data set.

Dataset can be made or downloaded from kaggle.com Assume we have a csv file named iris follows dataset with following example data.

	Sepal.length	Sepal.width	pital-length	pital width	Species
0	5.1	3.5	1.4	0.2	
1	4.9	3.0	1.4	0.2	
2	4.7	3.2	1.3	0.2	
3	4.6	3.1	1.5	0.2	

2. Importing libraries

```
import pandas as pd
import numpy as np.
```

```
import matplotlib . pyplot as plt.
from sklearn . preprocessing import Standard Scaler.
from sklearn . preprocessing import label encoder,
                                    ordinal Encoder.
```

3. Importing the Data set:-

```
df = pd . read - csv ('/content / Iris . csv'),
df . head ()
df . tail ().
```

4. finding the Missing Data.

```
df . isnull ().sum ().
```

5. Encoding Training and Testing Data:-

```
df _ le = df . copy ()
le = label Encoder ()
df - le ["species _ LE"] = le . fit . transform (df . le ["species"])
df . le [["species", "Species - LE"]].
```

6. Spitting Training and Testing Data:-

```
x = iris . data.
y - iris . target.
x _ train , x test , y - train . ,y train _ test - split.
        (x, y , test _ = 0.2 , random - state = 42.
```

7. feature scaling.

```
Scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)
print("Training data shape:", x_train.shape)
print("Testing data shape: ", x_test.shape)
```